TAEYOUNG YUN

Ph.D student @ KAIST

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RESEARCH INTERESTS

I aim to develop agents that make reliable decisions when training environments and reward signals imperfectly reflect real-world conditions and user goals. I am particularly interested in aligning training environments with intended tasks and learning policies that remain robust to uncertainty and changes in the environment.

My research in reinforcement learning and generative modeling provides a foundation for this goal through efficient exploration and reliable decision-making. I develop off-policy reinforcement learning methods to generate high-quality and diverse candidates, and generative model-based methods for optimization in complex search spaces. My work spans LLM post-training, text-to-image generation, biological sequence design, and simulation-based optimization, including traffic control and infrastructure layout design.

EDUCATION

03/2024 - Current **Ph.D Student in Industrial and Systems Engineering** ***KAIST***

Supervised by Jinkyoo Park

08/2022 - 02/2024 **M.S in Graduate School of AI** ***KAIST***

Supervised by Jinkyoo Park
M.S Thesis: Offline Meta Black-box Optimization Framework for Intelligent Traffic Light Management System

03/2018 - 08/2022 **B.S in Industrial and Systems Engineering & Computer Science** ***KAIST***

INTERNSHIPS

08/2026 - 10/2026 **Research Intern at NaverLabs Europe** ***Grenoble, France***

Hosted by Darko Drakulic and Sofia Michel
Develop a scalable RL algorithm for online multi-agent coordination under uncertainty

06/2025 - 11/2025 **Research Intern at Mila** ***Montreal, Canada***

Hosted by Yoshua Bengio
RL-based red-teaming with evolving environments by safety fine-tuning the victim LLM to promote easy-to-hard exploration.

09/2024 - 03/2025 **Visiting Intern at HKUST** ***Hong Kong (Remote)***

Hosted by Ling Pan
Fine-tuning LLM with GFlowNets to generate diverse and effective prompts for text-to-image diffusion models.

03/2021 - 08/2021 **Research Intern at Kakao Recommendation Team** ***Seoul, Korea***

Develop contextual bandit algorithms for a personal recommendation.
Analyze the gap between simulation and real-world deployment.

SELECTED PUBLICATIONS

*: Equal Contribution

NeurIPS, 2026 (based on ICMLW)
Automated Kernel Discovery for Understanding High-dimensional Bayesian Optimization
Taeyoung Yun*, Woocheol Shin*, Inhyuck Song, Jaewoo Lee, and Jinkyoo Park

NeurIPS, 2026 (based on NIPSW)
Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization
Kiyoung Om*, Kyuil Sim*, **Taeyoung Yun***, Hyeongyu Kang, and Jinkyoo Park

ICML, 2026
Active Attacks: Red-teaming LLMs via Adaptive Environments
Taeyoung Yun, Pierre-Luc St-Charles, Jinkyoo Park, Yoshua Bengio, and Minsu Kim

ICML, 2025 (based on ICLRW)
Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization
Taeyoung Yun*, Kiyoung Om*, Jaewoo Lee, Sujin Yun, and Jinkyoo Park

CVPR, 2025
Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation
Taeyoung Yun, Dinghuai Zhang, Jinkyoo Park, and Ling Pan

NeurIPS, 2024
Guided Trajectory Generation with Diffusion Models for Offline Model-based Optimization
Taeyoung Yun, Sujin Yun, Jaewoo Lee, and Jinkyoo Park

HONORS & AWARDS

2025 **Qualcomm Innovation Fellowship Finalist** ***Seoul, Korea***
Paper: Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation

2025 **Top Reviewer** ***NeurIPS***
Honor for Top 8% Reviewers

2021 **Dean's List** ***KAIST***
Honor for Top 2% Students

2021 **Excellence Award (2nd Place)** ***Seoul, Korea***
Big Data Competition Hosted by NH

TALKS & PRESENTATIONS

2026 **NaverLabs Europe**
Train online RL policy for multi-agent coordination

2026 **AutoML Seminar**
Posterior Inference in Generative Models for High-dimensional Black-box Optimization [Video]

INDUSTRIAL PROJECTS

03/2026 - 08/2026 **Agentic AI for Traffic-related Question Answering** ***Seoul, Korea***

Collaborate with GS

Develop an Agentic AI system for developing LLM-based for better traffic-related question answering framework.

09/2024 - 09/2025 **Traffic Network Layout Optimization** ***Seoul, Korea***

Collaborate with GS

Develop a Generative model-based design algorithm for optimizing traffic network layout on a given traffic pattern.

03/2023 - 03/2024 **Incentive Design for Managing Taxi Fleet** ***Daejeon, Korea***

Collaborate with ETRI

Develop an RL-based incentive design algorithm for rebalancing taxi fleets to resolve the taxi imbalance problem.

03/2022 - 03/2023 **Traffic Light Optimization** ***Seoul, Korea***

Collaborate with KT

Develop a Bayesian optimization algorithm for managing multiple traffic lights in the real world to reduce congestion.

FULL PUBLICATIONS

Preprints

Arxiv, 2026 **[P4] Noise-Space Reward Alignment for Few-Step Generative Models**
Woocheol Shin, **Taeyoung Yun**, Jiwon Sung, Hyeongyu Kang, Jeongmin Park, Jaewoo Lee, Kyuil Sim, Dohyun Kim, Yutong He, and Jinkyoo Park
[Paper] / [Code]

Arxiv, 2026 **[P3] Decoupling Exploration and Exploitation in GFlowNets via Policy Evaluation**
Taeyoung Yun*, Inhyuck Song*, Jinkyoo Park, and Ling Pan
[Paper] / [Code]

Arxiv, 2026 (based on ICMLW) **[P2] Robust Exploration through Generative Replay**
Inhyuck Song, **Taeyoung Yun**, Jaewoo Lee, and Jinkyoo Park
[Paper] / [Code]

Arxiv, 2025 **[P1] Improving Sampling Distribution of Off-policy Training in Generative Flow Networks**
Taeyoung Yun, Sujin Yun, Jinkyoo Park, and Ling Pan
[Paper] / [Code]

Conference Publications

NeurIPS, 2026 (based on ICMLW) **[C15] Automated Kernel Discovery for Understanding High-dimensional Bayesian Optimization**
Taeyoung Yun*, Woocheol Shin*, Inhyuck Song, Jaewoo Lee, and Jinkyoo Park
[Paper] / [Code]

NeurIPS, 2026 (based on ICMLW) **[C16] Aligning Few-Step Generative Models by Amortizing Sample-based Variational Inference**
Jaewoo Lee*, Hyeongyu Kang*, Dohyun Kim, Kyuil Sim, Woocheol Shin, Minsu Kim, **Taeyoung Yun**, Jeongjae Lee, Sanghyeok Choi, Tabitha Edith Lee, Jongchul Ye, Jinkyoo Park
[Paper] / [Code]

NeurIPS, 2026 (based on NIPSW)
[C17] Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization
Kiyoung Om*, Kyuil Sim*, **Taeyoung Yun***, Hyeongyu Kang, and Jinkyoo Park
[Paper] / [Code]

CIKM, 2026 (Short Paper)
[C14] SUMO-SimQA: A Benchmark for LLM Cross-File Reasoning over Traffic Simulation Files
Jeeyun Chang*, **Taeyoung Yun***, Minwoo Jeong, Jinkyoo Park, and Yoonjin Yoon
[Paper] / [Code]

ICML, 2026
[C13] Active Attacks: Red-teaming LLMs via Adaptive Environments
Taeyoung Yun, Pierre-Luc St-Charles, Jinkyoo Park, Yoshua Bengio, and Minsu Kim
[Paper] / [Code]

ICLR, 2026
[C12] Diffusion Alignment as Variational Expectation-Maximization
Jaewoo Lee, Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, Woocheol Shin, **Taeyoung Yun**, Kiyoung Om, and Jinkyoo Park
[Paper] / [Code]

WSDM, 2026
[C11] Urban Traffic Network Layout Optimization with Guided Discrete Diffusion Models
Taeyoung Yun, Inhyuck Song, Woocheol Shin, Yujin Shin, Sungpil Woo, Sunhwan Lim, and Jinkyoo Park
[Paper] / [Code]

KDD, 2025
[C10] Wind Farm Layout Optimization with Diffusion Models
Yujin Shin*, **Taeyoung Yun***, Sujin Yun, Sungpil Woo, Sunhwan Lim, and Jinkyoo Park
[Paper] / [Code]

ICML, 2025 (based on ICLRW)
[C9] Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization
Taeyoung Yun*, Kiyoung Om*, Jaewoo Lee, Sujin Yun, and Jinkyoo Park
[Paper] / [Code]

ICML, 2025 (based on NIPSW)
[C8] Improved Off-Policy Reinforcement Learning in Biological Sequence Design
Hyeonah Kim, Minsu Kim, **Taeyoung Yun**, Sanghyeok Choi, Emmanuel Bengio, Alex Hernández Garcia, and Jinkyoo Park
[Paper] / [Code]

CVPR, 2025
[C7] Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation
Taeyoung Yun, Dinghuai Zhang, Jinkyoo Park, and Ling Pan
[Paper] / [Code]

ICLR, 2025
[C6] Adaptive Teachers for Amortized Samplers
Minsu Kim*, Sanghyeok Choi*, **Taeyoung Yun**, Emmanuel Bengio, Leo Feng, Jarrid Rector-Brooks, Sungsoo Ahn, Jinkyoo Park, Nikolay Malkin, and Yoshua Bengio
[Paper] / [Code]

NeurIPS, 2024
[C5] Guided Trajectory Generation with Diffusion Models for Offline Model-based Optimization
Taeyoung Yun, Sujin Yun, Jaewoo Lee, and Jinkyoo Park
[Paper] / [Code]

NeurIPS, 2024 (based on ICLRW)
[C4] GTA: Generative Trajetory Augmentation with Guidance for Offline Reinforcement Learning
Jaewoo Lee*, Sujin Yun*, **Taeyoung Yun**, and Jinkyoo Park
[Paper] / [Code]

KDD, 2024
[C3] An Offline Meta Black-box Optimization Framework for Adaptive Design of Urban Traffic Light Management Systems
Taeyoung Yun*, Kanghoon Lee*, Sujin Yun, Ilmyung Kim, Won-Woo Jung, Min-Cheol Kwon, Kyujin Choi, Yoohyeon Lee, and Jinkyoo Park
[Paper] / [Code]

ICML, 2024
(based on NIPSW)

[C2] Learning to Scale Logits for Temperature-conditional GFlowNets
Minsu Kim*, Juhwan Ko*, **Taeyoung Yun***, Dinghuai Zhang, Ling Pan, Woochang Kim, Jinkyoo Park, and Yoshua Bengio
[Paper] / [Code]

ICLR, 2024
(Spotlight)

[C1] Local Search GFlowNets
Minsu Kim, **Taeyoung Yun**, Emmanuel Bengio, Dinghuai Zhang, Yoshua Bengio, Sungsoo Ahn, and Jinkyoo Park
[Paper] / [Code]

Journal Publications

TMLR, 2026
(Survey Certification)

[J1] Offline Model-Based Optimization: Comprehensive Review
Minsu Kim, Jiayao Gu, Ye Yuan, **Taeyoung Yun**, Zixuan Liu, Yoshua Bengio, Can Chen
[Paper] / [Code]

ACADEMIC SERVICES

2026 **Reviewer**

AAAI, KDD, ICLR, AAMAS, AISTATS, TMLR, TPAMI, ICML, RLC, NeurIPS

2025 **Reviewer**

ICLR, AAMAS, AISTATS, ICML, KDD, TMLR, NeurIPS

TEACHING EXPERIENCES

2026 **Teaching Assistant** ***KAIST***

IE579: Game Theory and Multi-Agent Reinforcement Learning

2025 **Teaching Assistant** ***KAIST***

IE343: Statistical Machine Learning
IE437: Data-Driven Decision Making and Control

2024 **Teaching Assistant** ***KAIST***

IE437: Data-Driven Decision Making and Control
IE481: Manufacturing & Artificial Intelligence

2022 **Teaching Assistant** ***KAIST***

MAS480: Introduction to Scientific Machine Learning